

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA
Course: Web Technologies II

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a web server? How does HTTPS ensure secure communication between clients and servers? Provide a brief explanation of the HTTP request and response mechanism. 8
- b) What are arrays? What are the different ways of creating an array in PHP? Explain with suitable example. 7
2. a) Write a PHP code to validate and insert doctor's information into the doctors table (doctor_name, specialization, phone_number, hospital_name) in a database hospitaldb. (An HTML form is must.) 7

OR

What are Prepared Statements in PHP? How do prepared statements help prevent SQL Injection? Explain with suitable example.

- b) Explain the role of cookies and sessions in maintaining state between a client and a server. Write a simple PHP program to demonstrate the use of cookies in PHP. 8
3. a) What is a Servlet? Explain how a session is handled using a Servlet with a suitable example. 7

OR

Design a "Contact Us" HTML form for Pokhara University with fields (Name, Email, Phone Number, Subject, Message) and use a Servlet to display the submitted data using the POST method.

- b) What is JSP? Explain its lifecycle, including all the methods involved, with a suitable example 8
4. a) Write a JSP program using JDBC to display product name, category, and price of those products which belong to the "Electronics" category and whose price is greater than Rs. 5,000. (Make your own assumptions for table and database.). 7

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| b) | What is the MVC (Model-View-Controller) design pattern? How does it contribute to the development of web applications? Provide an explanation of two frameworks that follow the MVC pattern. | 8 |
| 5. a) | What are the uses of CMS? Explain popular CMS used in the market today for digital content management. | 7 |
| b) | Define Web Service. Create a simple RESTful web service endpoint for Create and Read all record of the student's object having (id, name, email and gpa) as an attribute. | 8 |
| 6. a) | What do you mean by Vulnerabilities, Threats, Attacks in web application? How do you protect from them? | 7 |
| b) | What do you mean by Web Application Security? Explain about the best security practices for the web development. | 8 |
| 7. | Write short notes on: (Any two) | 2×5 |
| a) | Sending Email in PHP | |
| b) | Java Servlet | |
| c) | SOAP | |